

Submitted by:-

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AICT

Assignment # 2

Number System Conversion:-

Binary:-

Binary $\rightarrow$ Decimal:-

Binary number is $= 101101_2$

			Result
1	2^5	32×1	32
0	2^4	16×0	0
1	2^3	8×1	8
1	2^2	4×1	4
0	2^1	2×0	0
1	2^0	1×1	1

Now add the result:

$$32 + 0 + 8 + 4 + 0 + 1 = 45$$

Decimal = 45_{10}

Binary $\rightarrow$ Octal:-

Binary = 101101

make Group

101 101

Binary	Octal value
101	5
101	5

Octal value = 55_8

Submitted to:-

Dr. Mrs. Zaheer
Mussain Fida

Binary $\rightarrow$ Hexadecimal

Binary = 101101

Binary	Hexadecimal
0010	2
1101	D

Hexadecimal = 2D₁₆

Decimal:-

Decimal $\rightarrow$ Binary:-

Decimal = 25₁₀

2	25
2	12 - 1
2	6 - 0
2	3 - 0
1	1 - 1

Binary = 11001₂

Decimal $\rightarrow$ Octal

8	25
	3 - 1

Octal = 31₈

Decimal $\rightarrow$ Hexadecimal

16	25
	1 - 9

Hexadecimal = 19₁₆

Octal:-

Octal $\rightarrow$ Binary:-

Each octal digits = 3 binary bits

$$\text{Octal} = (57)_8$$

$$5 = 101$$

$$7 = 111$$

$$\text{Binary} = (101111)_2$$

Octal $\rightarrow$ Decimal:-

$$\text{Octal} = (57)_8$$

$$= 5 \times 8^1 + 7 \times 8^0$$

$$= 40 + 7$$

$$\text{Decimal} = (47)_{10}$$

Octal $\rightarrow$ Hexadecimal:-

$$\text{Octal} = (57)_8$$

First convert into Binary:

$$57 = 101111$$

$\div$ in 4 bits

$$0010 \quad 1111$$

$$0010 = 2$$

$$1111 = F$$

$$\text{Hexadecimal} = 2F_{16}$$

Hexadecimal:-

Hexadecimal $\rightarrow$ Binary:-

$$\text{Hexadecimal} = (3A)_{16}$$

$$3 = 0011$$

$$A = 1010$$

$$\text{Binary} = (00111010)_2$$

Hexadecimal - Octal:-

$$\text{Hexadecimal} = (3A)_{16}$$

convert into Binary

$$(3A)_{16} \rightarrow (00111010)_2$$

$$000 \quad 111 \quad 010$$

$$000 = 0$$

$$111 = 7$$

$$010 = 2$$

$$\text{Octal} = (72)_8$$

Hexadecimal $\rightarrow$ Decimal:-

$$\text{Hexadecimal} = (3A)_{16} \quad \text{Here } A = 10$$

$$= (3) \times (16)^1 + (10) \times (16)^0$$

$$= 48 + 10$$

$$\text{Decimal} = (58)_{10}$$